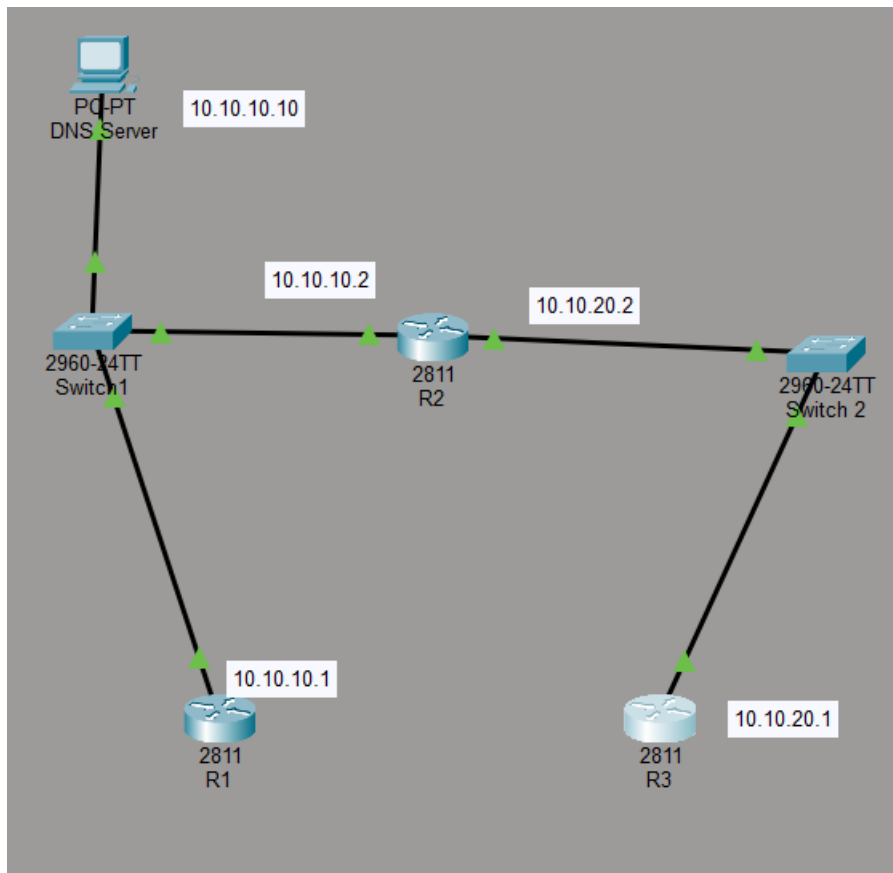


Week 3 practicals

16.) Screenshot of the topology



17.) This is me testing the name resolution from a router by pinging a hostname instead of an IP address.

```
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.1
Router(config)#ip domain-name lab.local
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 5/12/23 ms

Router#
```

18 and 19).I cleared the ARP cache on the router , then I examined the ARP cache on the routers with 'show arp' and identified the newly learned entry.

```
Router#clear arp-cache
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 - 00D0.97DE.5701 ARPA FastEthernet0/0
Internet 10.10.10.2 0 00D0.BC99.4901 ARPA FastEthernet0/0
Router#
```

20.) An ARP entry gives you the actual hardware (MAC) address of a host on the local subnet. A routing table entry only tells you the logical path (which network or next hop to send packets to).

21.) The routing table gives the logical path, but ARP provides the physical address needed for the packet to leave the router's interface and reach the next hop.